Self-Consistent 2-D MOSFET Surface Potential Derivation

Nomenclature (First Page)

x: vertical silicon coordinate
y: channel coordinate
L: channel length
tox: oxide thickness
psi(x,y): electrostatic potential
phi_s(y): surface potential
VG: gate voltage
VDS: drain voltage
Vbi: built-in potential
NA, ND: doping concentrations
lambda: electrostatic scaling length

Derivation summary

```
Poisson equation:
nabla^2 psi(x,y) = -rho/epsilon_s

rho = q(p-n-N_A)

Surface potential:
phi_s(y)=psi(0,y)

d2(phi_s)/dy2=(phi_s-VON)/lambda^2

lambda=sqrt(eps_s/eps_ox * tox * Xdmax)

phi_s(y)=VON+(Vbi-VON)sinh((L-y)/lambda)/sinh(L/lambda)
+(Vbi+VDS-VON)sinh(y/lambda)/sinh(L/lambda)

The attached Python code generates 2D contour,
surface potential, and 3D plots.
```